Computer Networks Lab
Week 03

Capturing HTTP and HTTPs Traffic using Wireshark

Tools:

Wireshark: Download from https://www.wireshark.org/download.html
Web Browser: Chrome/Edge
Text Editor/IDE: VS Code, Notepad++ or any editor to modify HTML

Objectives:

1. Install and configure Wireshark to capture network traffic.
2. Create and use an HTML form to generate HTTP traffic with sensitive data (e.g., username, password, age).
3. Capture and analyze HTTP traffic to observe plaintext data vulnerabilities.
4. Browse a secure website (e.g., Wikipedia) to generate and capture HTTPS traffic, observing encrypted data.
5. Compare HTTP and HTTPS packet details to understand the role of SSL/TLS encryption.
6. Relate findings to real-world applications, such as secure data transmission for web services.

Introduction to Secure Digital Transmissions

In modern computer networks, the security of data transmissions is crucial. Data is transmitted as digital signals over network cables or wireless channels. The efficiency of this transmission depends on multiple factors, including transmission modes, encoding schemes, bandwidth usage, data rate, signal rate, and error detection mechanisms. These elements collectively determine how efficiently and accurately data moves across a network. As users interact with websites, they frequently send and receive sensitive information, such as login credentials, payment details, and personal data. If this information is transmitted insecurely, attackers can intercept and misuse it, leading to data breaches, identity theft, and other cybersecurity threats.

Two widely used protocols for transmitting data over the web are:

- **HTTP (Hypertext Transfer Protocol):** A protocol for transferring web data (e.g., web pages, forms) between clients (browsers) and servers. It operates on port 80 and sends data in **plaintext**, making it vulnerable to interception.
- **HTTPS (Hypertext Transfer Protocol Secure):** An extension of HTTP that uses **SSL/TLS** for encryption, ensuring data confidentiality, integrity, and authenticity. It operates on port 443, protecting sensitive data (e.g., passwords, credit card details).

- **Wireshark:** An open-source tool for capturing and analyzing network packets, allowing users to inspect protocol details (e.g., HTTP POST requests, HTTPS encrypted data).

- **Packet:** A unit of data transmitted over a network, containing headers (metadata) and payload (actual data).

- **SSL/TLS (Secure Sockets Layer/Transport Layer Security):** Cryptographic protocols that encrypt data, verify server identity, and prevent tampering. TLS is the modern version, used in HTTPS.

- **DNS (Domain Name System):** A system that translates domain names (e.g., google.com) to IP addresses (e.g., 216.58.216.164). Relevant for resolving website addresses in this lab.

This lab will guide you through capturing and analyzing HTTP and HTTPS traffic using **Wireshark**, a popular network protocol analyzer. By submitting form data over HTTP and HTTPS, you will observe the critical differences between plaintext and encrypted transmissions.

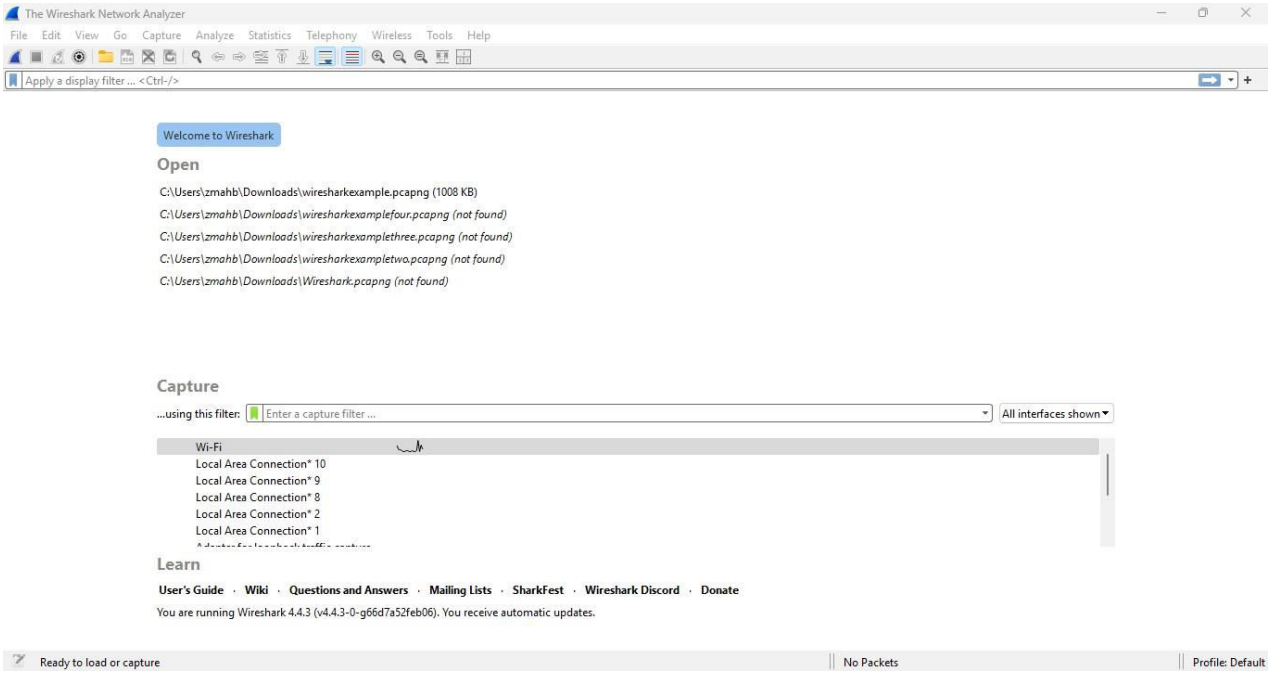
Capturing HTTP Traffic using Wireshark

To analyze HTTP traffic, we will use a simple HTML form provided in Google Classroom. This form contains only a **username** and **password** field and submits data using the HTTP protocol. Since HTTP transmits data in plaintext, we will use Wireshark to capture the network traffic and examine how easily sensitive information can be intercepted.

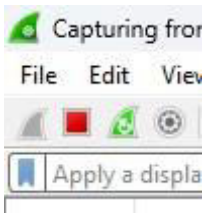
Start by opening **Wireshark** and selecting the appropriate network interface, either **Wi-Fi** or **Ethernet**, depending on how your computer is connected to the internet. Double click on the given options to begin the packet capture. Once started you can stop by clicking the red button next to the blue fin, or restart by clicking on the **Start Capture** button (the blue fin button). Once Wireshark is recording network traffic, open the HTML form in your web browser and enter test credentials for the **username** and **password** fields. Click the **Submit** button to send the form data over the network.

After submitting the form, stop the packet capture in Wireshark. To filter out irrelevant traffic, enter *http* in the filter bar and press **Enter**. This will display only HTTP packets. Look for a **POST request** in the **Info** column, as this is the request that carries the form data. Click on the **POST request** packet to examine its details. In the **Packet Details pane**, expand the **Hypertext**

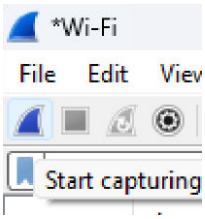
Transfer Protocol (HTTP) section, followed by the **HTML Form URL Encoded** section.
Here, you will be able to see the **username** and **password** in plaintext, exactly as they were entered in the form.
This confirms the fundamental security flaw of HTTP: all transmitted data is unencrypted, meaning that anyone monitoring the network can intercept and read sensitive information such as login credentials. You can also observe some screenshots related to the steps described above, in the sections below.



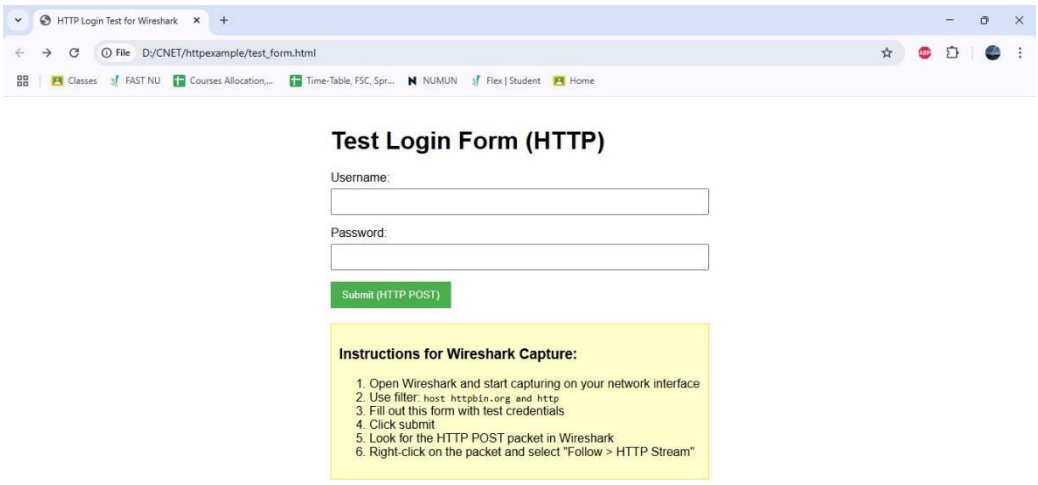
Selecting Wi-Fi interface for capturing packets.



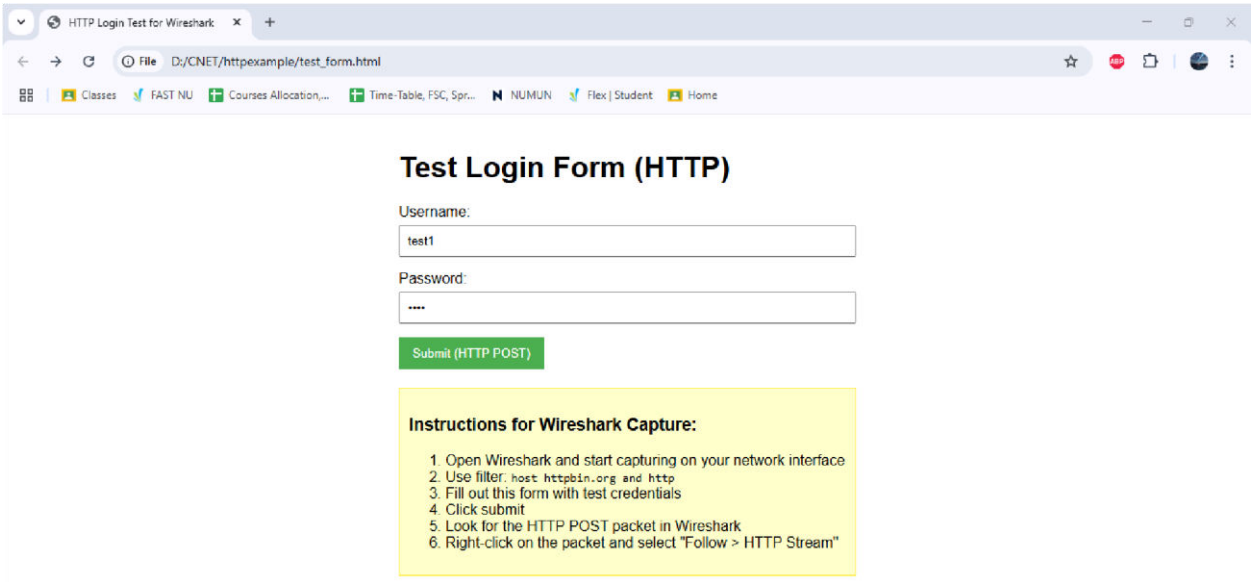
Stop button



Start button

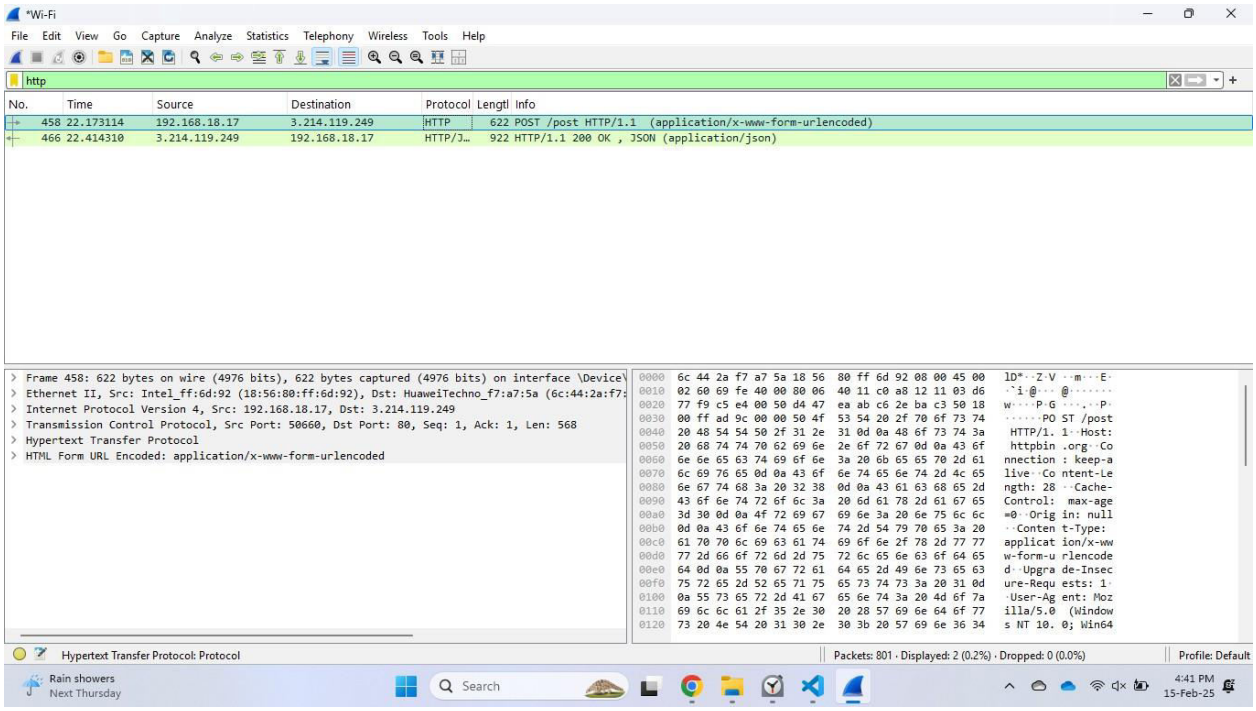


Accessing the website with the HTML form

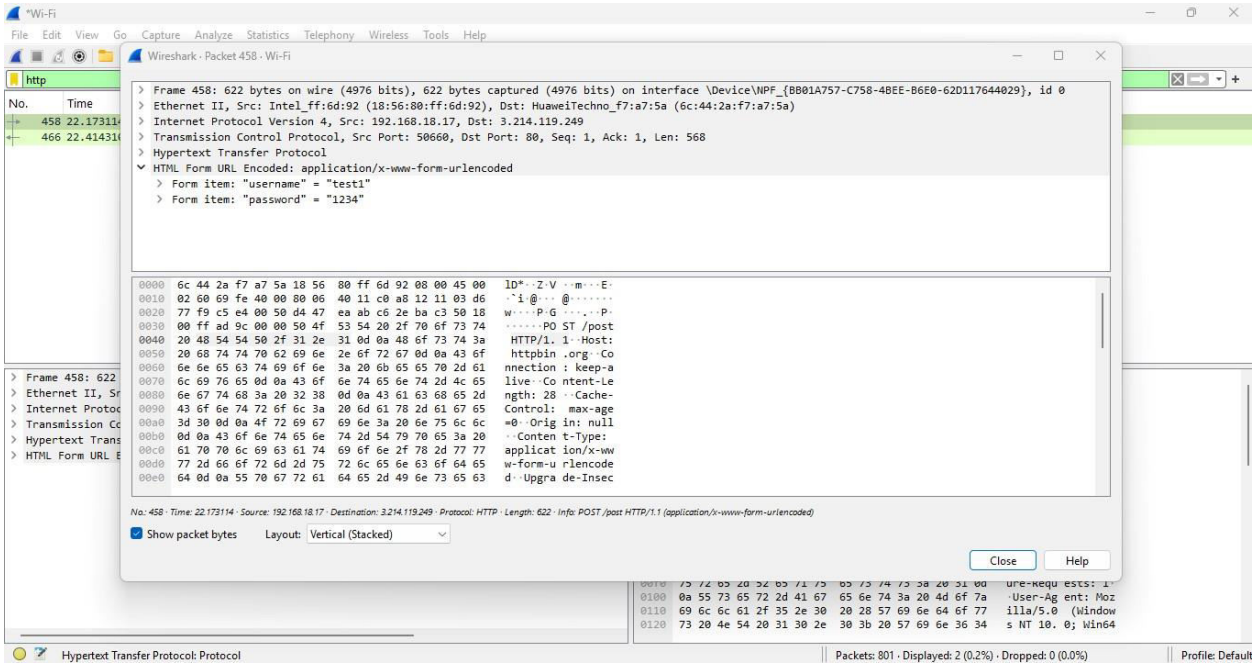


Entering insecure credentials

Response page after insecure credentials have been entered



Filtering packets to show HTTP traffic



Opening relevant packet to see details sent through HTTP request

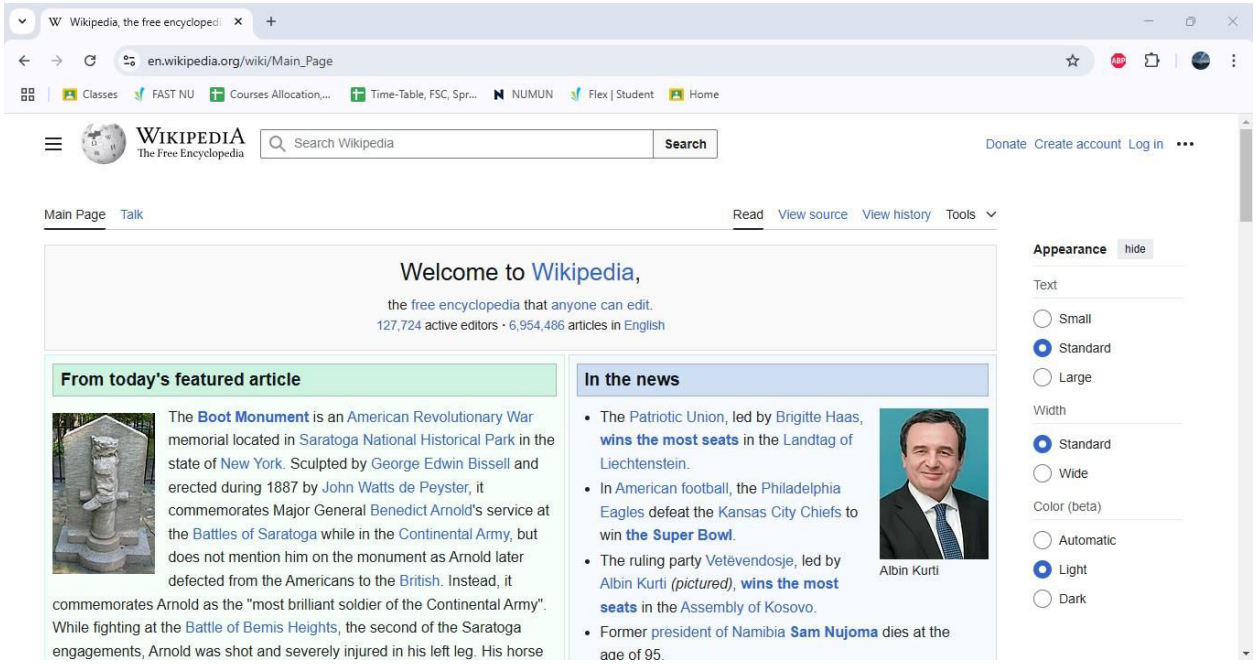
Capturing HTTPS Traffic using Wireshark

To analyze HTTPS traffic, we will visit **Wikipedia** and search for the term **"cheese"**, then browse through different pages on the website. Unlike HTTP, HTTPS encrypts all transmitted data using SSL/TLS, making it unreadable even when intercepted. Our goal is to capture HTTPS packets in Wireshark and observe how the encryption prevents us from seeing the actual data being transmitted.

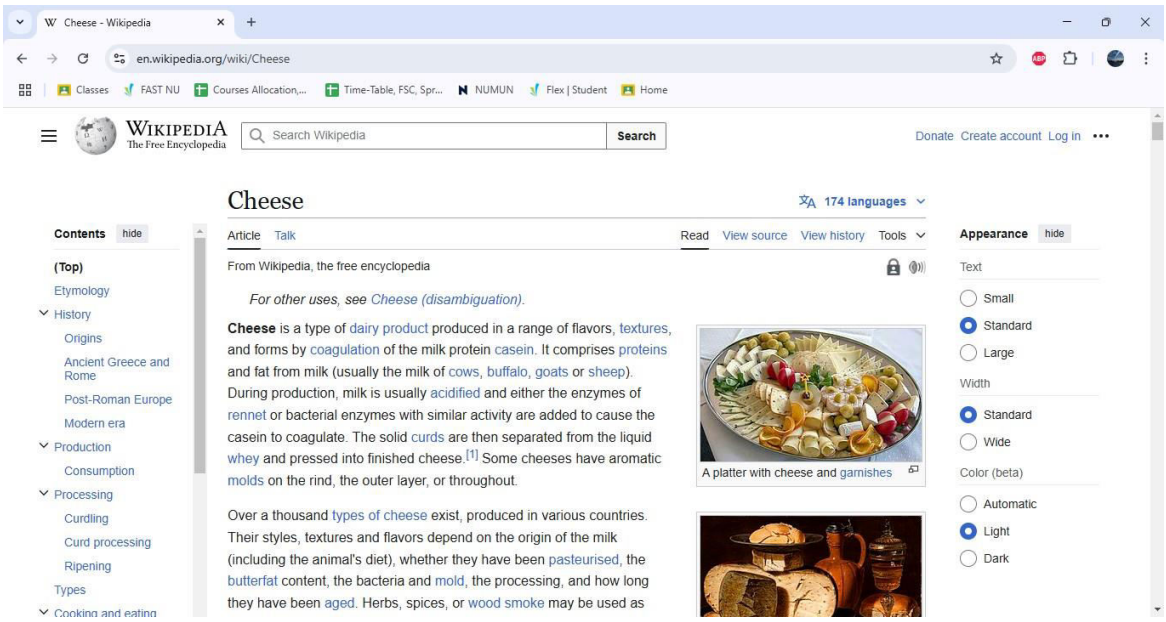
Begin by opening **Wireshark** and selecting the appropriate network interface. Click on the **Start Capture** button to begin recording network traffic. With Wireshark running in the background, open a web browser and navigate to **https://www.wikipedia.org/**. In the Wikipedia search bar, type **"cheese"** and press **Enter**. Browse through various cheese-related articles for a minute to generate sufficient network traffic.

Once you have interacted with the website, stop the packet capture in Wireshark. To isolate HTTPS traffic, apply a filter by entering either *tcp.port == 443* or *ssl* in the filter bar and pressing **Enter**. Unlike HTTP, you will not see clearly readable text in the captured packets. Instead, locate a packet labeled **Application Data** under the **TLSv1.2** or **TLSv1.3** protocol. Clicking on this packet will reveal an **Encrypted Application Data** section in the **Packet Details pane**, indicating that the actual content of the transmission is securely encrypted.

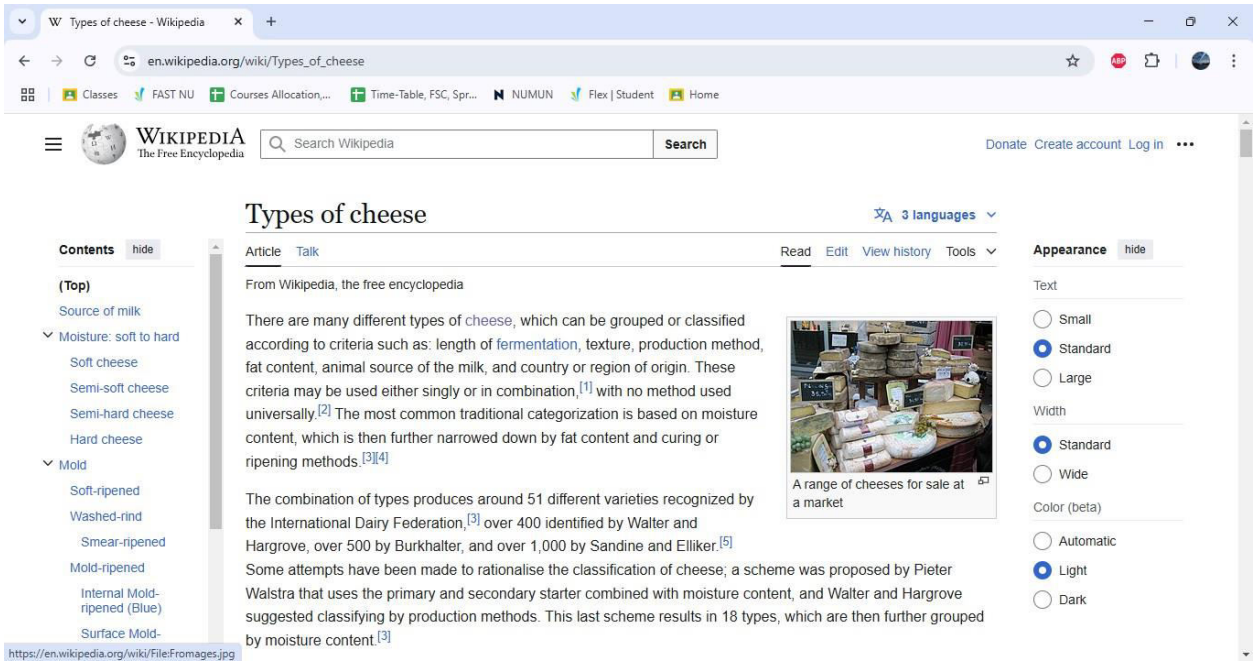
This confirms that HTTPS successfully protects sensitive information by encrypting it before transmission. Unlike HTTP, where data is sent in plaintext, HTTPS ensures that even if network traffic is intercepted, the actual content remains hidden from unauthorized parties. This encryption is what makes HTTPS the preferred choice for secure digital communications, safeguarding user privacy and protecting sensitive data from cyber threats. You can also observe some screenshots related to the steps described above, in the sections below.



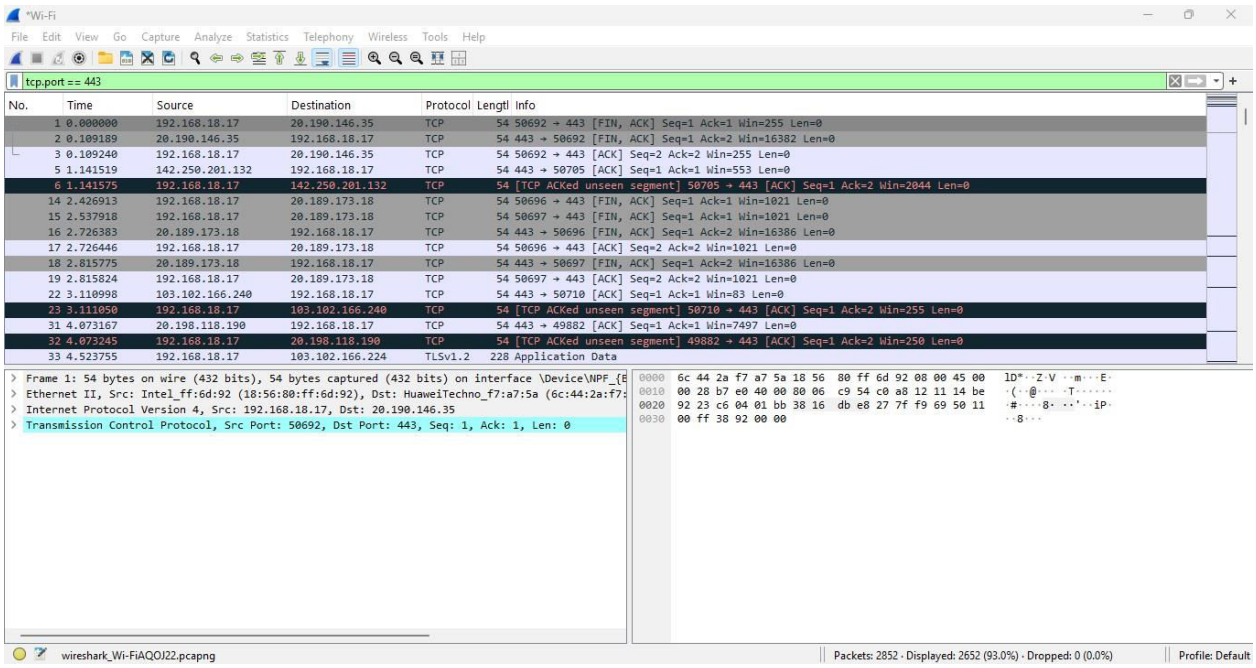
Going to Wikipedia while Wireshark runs in the background



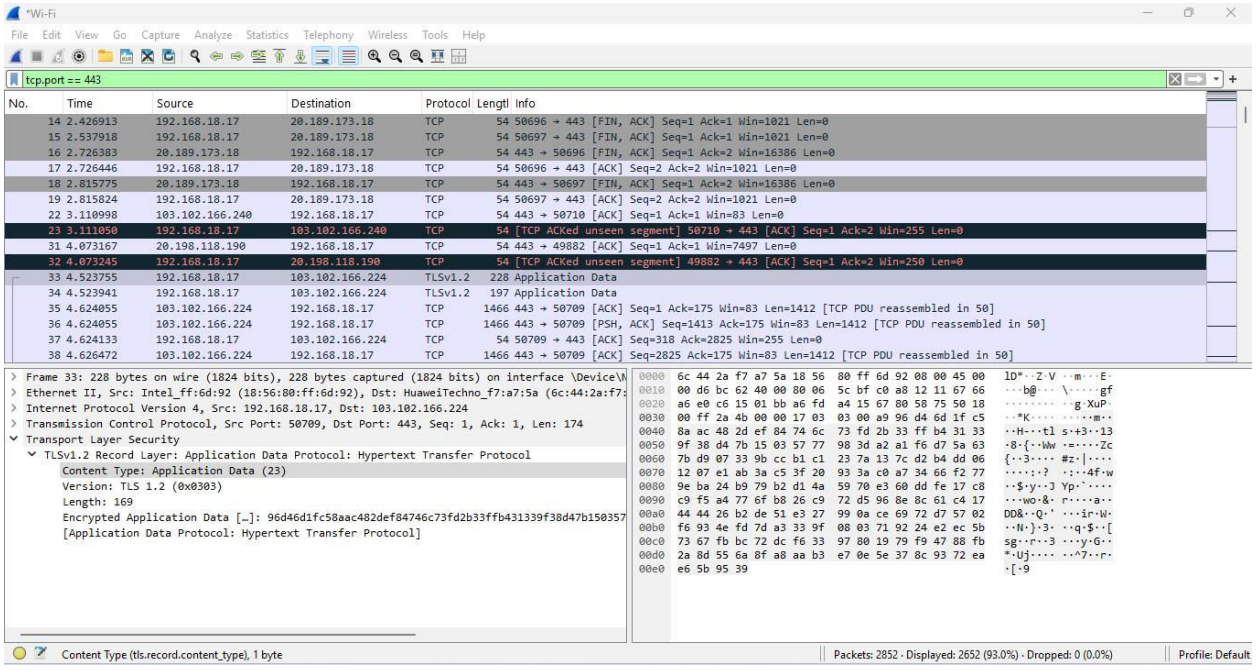
Results after searching “cheese”



Browsing through different cheese related pages.



Applying tcp.port == 443 (the port used for HTTPS traffic) to see HTTPS traffic



Observing encrypted application data

Lab Tasks

Task 1: Analyzing HTTP Traffic (10 marks)

Using the code given in the google classroom, you will generate HTTP traffic and show insecurely sent data clearly displayed on Wireshark. For this, download the code from the GCR and, using any IDE, add a field for ***age*** in the form and add your **roll number and name** in place of <h1> </h1>. Then you will follow these steps, recording each step by taking screenshots of it and attaching them to a pdf file/word file:

Start the Wireshark Capture

1. Open **Wireshark**.
2. Select the **network interface** that your computer uses for internet access (e.g., Wi-Fi or Ethernet).
3. Click **Start Capture**.

Submit the Form

4. Open ***test_form.html*** in a web browser.
5. Enter the following details in the form:
    - **Username:** Your roll number
    - **Password:** Any random password (e.g; 1234)
    - **Age:** Any number
6. Click **Submit**.

Filter and Analyze HTTP Packets

7. Stop the Wireshark capture.
8. In the **Wireshark Filter Bar**, type *http* and press **Enter** to isolate HTTP traffic.
9. Locate the **POST** request packet that contains the submitted form data.
10. Click on the packet and expand the **HTML Form URL Encoded** section in the lower panel.
11. Take a **screenshot** showing the captured username, password, and age in plaintext.

Also, **answer** the following question:

Q1. Why is HTTP considered insecure for transmitting sensitive data? Explain the risks of plaintext transmission and how attackers can exploit it.

Task 2: Analyzing HTTPS Traffic (10 marks)

By navigating through different pages on Wikipedia, you will generate HTTPS traffic that you will then capture and analyze to see how encryption works. For this, you will go to https://www.wikipedia.org/ and browse through several pages related to tomatoes. The steps you will follow, recording each step by taking screenshots of it and attaching them to a pdf file/word file, are:

Start a Wireshark Capture

1. Open **Wireshark** and start a new packet capture session (details also given in task 1).

Visit Wikipedia and look for a page related to Tomatoes

2. Open your web browser and navigate to **Wikipedia's Tomato page**.
3. Browse the page for up to 1 minute to generate network traffic.

Filtering and Analyzing HTTPS Packets

4. Stop the Wireshark capture.
5. In the **Wireshark Filter Bar**, type *tcp.port == 443* or *ssl* and press **Enter** to isolate HTTPS traffic.
6. Locate a packet labeled **Application Data**—this contains encrypted content.
7. Take a **screenshot** of the packet details showing that the data is encrypted.

Also, **answer** the following questions:

Q1. What differences do you notice between HTTP and HTTPS traffic? Compare the packet details for HTTP vs. HTTPS and note how the data is secured.

Q2. Why is HTTPS considered more secure than HTTP? Explain the role of SSL/TLS encryption in preventing data interception and tampering.

Task 3: Capturing and Understanding Packet Transmission (10 marks)

The first step in analyzing digital transmission is to capture network packets and examine their transmission characteristics. You will capture **ICMP (ping) packets** to observe how data is transmitted serially over the network.

For this you will:

Open **Wireshark** and select the **active network interface** (Ethernet or Wi-Fi). Start a packet

capture session by clicking the **Start Capture button**. Open the command prompt on your system and run this: ping -n 5 www.google.com . Which will send five **ICMP echo request** packets to Google's server. After the ping operation completes, **stop the packet capture** in Wireshark. In the Wireshark **filter bar**, enter the icmp filter to isolate ICMP packets.

Select an **ICMP request packet** and examine the **Frame Details**. Record the following information on a document:

1. Packet arrival time (timestamp) [any packet]
2. Frame length (bytes) [any packet]
3. Time difference between consecutive packets (inter-packet delay) [first and last request]

Based off this information, answer the following questions:

Q1. How frequently are packets being sent?

Q2. What is the **average packet size**, and how does it compare to other traffic types? (hint: you will have to generate different traffic)

Q3. What transmission mode is used?

Task 4: Analysing DNS Traffic (10 marks)

Capture and analyze DNS traffic to understand how domain names (e.g., www.google.com) resolve to IP addresses (e.g., 142.250.202.238).

1. Open **Wireshark** select your interface and **start capturing** network traffic.
2. Open Command Prompt and run:

   nslookup wikipedia.org

3. Note the IP Address 103.102.166.224 and capture the screenshot as below:

```
C:\Users\amina>nslookup wikipedia.org
Server:  dns.google
Address:  8.8.8.8

DNS request timed out.
    timeout was 2 seconds.
Non-authoritative answer:
Name:    wikipedia.org
Addresses:  2001:df2:e500:ed1a::1
          103.102.166.224
```

4. Stop Wireshark capture.
5. Filter and Analyze DNS packets

- In the filter bar, type **dns** and press Enter.
- Find a DNS Query packet for www.wikipedia.org and a DNS Response packet with the IP.
- Click the response packet, expand Domain Name System (response) > Answers. Verify the IP (e.g., 103.102.166.224).
- **Screenshot**: Capture Wireshark showing the DNS response (label: "Figure: DNS Response Packet").

6. Answer the Question:
   Q1: Why is DNS critical for HTTP/HTTPS communication, and how does it relate to secure data transmission?

Submission Guidelines:

- Submit a single PDF document AND THE HTML PAGE containing all tasks to the GCR. (no wireshark files)
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **J_i221234_Lab1**. Rename the HTML file as **rollnumber_name.html**.
- In case a demo is conducted, there will be deductions based on your answers to the questions asked.

NOTE: You must show clear screenshots of every step you take. Marks will be deducted for this mistake and in some cases you may be awarded 0 (this applies to all tasks).

Marking Criteria (45 marks):

Either 0 [not achieved], half [partial achieved] or full [achieved]

Task 1: Analyzing HTTP Traffic (10 Marks)

1. **Modification of HTML Form (2 Marks)**
   - Successfully added the "Age" field in the form and also their name and roll number - ZERO OR HALF MAX FOR NOT DISPLAYING ROLL NUMBER/NAME.
2. **Wireshark Capture & HTTP Packet Analysis (3 Marks)**
   - Captured HTTP traffic correctly, filtered for HTTP packets, and located the POST request with form data.
3. **Screenshots (3 Marks)**
   - Provided clear, labeled screenshots of each step, including the captured username, password, and age in plaintext.
4. **Answer to Q1 (2 Marks)**

- Clearly explained why HTTP is insecure, the risks of plaintext transmission, and how attackers can exploit it.

Task 2: Analyzing HTTPS Traffic (10 Marks)

1. **Wireshark Capture & HTTPS Packet Analysis (3 Marks)**
   - Captured HTTPS traffic correctly, filtered for HTTPS packets, and located encrypted Application Data.
2. **Browsing Wikipedia Pages (2 Marks)**
   - Visited and browsed Wikipedia pages related to tomatoes to generate HTTPS traffic.
3. **Screenshots (3 Marks)**
   - Provided clear, labeled screenshots of each step, including encrypted Application Data.
4. **Answers to Q1 & Q2 (2 Marks)**
   - Explained the differences between HTTP and HTTPS, and the role of SSL/TLS in securing data.

Task 3: Capturing and Understanding Packet Transmission (10 Marks)

1. **Packet Capture & Filtering (4 Marks)**
   a. 2 mark for successfully capturing ICMP packets in Wireshark.
   b. 2 mark for applying the icmp filter and selecting a request packet.
2. **Data Recording (4 Marks)**
   a. 2 mark for correctly recording packet arrival time, frame length, and inter-packet delay.
   b. 2 mark for analyzing inter-packet delay trends.
3. **Answering Questions (2 Mark)**
   a. 2 mark for accurately answering Q1–Q3 based on observations.

Task 4: Analyzing DNS Traffic (10 Marks)

1. **Packet Capture & Filtering (4 Marks)**
   a. Capture and filter DNS traffic.
2. **Screenshots and captions (4 Marks)**
   a. Necessary Screenshots
3. **Answering Questions (2 Mark)**
   a. 2 marks for accurately answering Q1 based on observations.

Submission Guidelines (5 Marks)

1. **PDF Document Submission and HTML file (2 Marks) -** NO ZIP files.
   - Single, clear, and well-organized PDF document with all tasks and screenshots.
     - HTML file containing their roll number and name (inside the file).

2. **File Naming (1 Mark)**
    - PDF and form files correctly named as per the guidelines.
3. **Details on the First Page (1 Mark)**
    - Name, roll number, section and lab number included.
4. **Clear and Professional Presentation (1 Mark)**
    - Screenshots, labels, and descriptions are neat and easy to understand.

Demo:

- If student cannot answer at all -5 marks.
- If student can answer somewhat -2 marks. - Correct answers results in no deductions.

Sample screenshot: